

Final Project Proposal - Discord AI Tutor

1. Project Framing

Background & Objective: Consistent learning is often hindered by a lack of structure. This project aims to build an autonomous AI tutor operating directly on Discord. Unlike traditional reactive agents, this system is "stateless" and relies on an event-driven architecture managed by scheduled tasks.

Workflow: The system operates in "Planning Mode" to generate syllabuses (either by reading provided PDFs or discovering topics from scratch via the web) and "Cron-Triggered Mode" to deliver daily micro-learning sessions.

Identity & Boundaries: Using OpenClaw, the tutor operates within strict boundaries, enforcing "Quiet Hours" (23:00 - 08:00 KST) and autonomously handling the complete content lifecycle, including the automated cleanup of source files and cron jobs upon track completion.

2. MCP Plan

1. Free Servers

- filesystem: Reads local course PDFs and writes cleanup deletions to the inbound folder to manage disk space.
- fetch: Reads external web pages to gather discovery-mode learning materials without needing paid search APIs.
- cron: Schedules native time-based triggers to deliver daily lessons and execute end-of-track cleanup precisely on time.

2. Custom Server

Name: TrackManager (Working prototype active)

Signatures & Input Schemas:

- add_new_track(user_id: string, mode: string, syllabus: string)
- get_active_track(track_id: string)
- increment_track_day(track_id: string)

3. Threat Model

- filesystem: Reads /media/inbound, writes deletions to /media/inbound. Credentials: None. An injected description could attempt to delete unintended local files outside the scope if not strictly sandboxed.
- fetch: Reads external domains, writes: None. Credentials: None. An injected description could attempt SSRF or prompt injection via malicious web text.
- cron: Reads schedule config, writes triggers. Credentials: None. An injected description could attempt a denial of service by scheduling infinite loops or bypassing quiet hours.
- Track Manager: Reads and writes strictly to TUTOR_MEMORY.md. Credentials: Discord API keys stored locally in .env. An injected description could attempt to corrupt syllabus states, manipulate day counters, or spoof user IDs.

4. First Log Entry

026-06-03 15:19:41.838089: add_new_track | Creating a DÉCOUVERTE track for Japanese cuisine for Michigan (517754371052797953).